

URL(Uniform Resource Locator) breakdown

`https://pupperazzi.com/dogs/45/profile?energy=high#photos`

- address that tells a browser or API client where to go and what to request.

- **Scheme / Protocol:**

`https://` → Uses secure HTTP for requests.

- **Domain / Host:**

`pupperazzi.com` → The website's main address.

- **Path:**

`/dogs/45/profile` → Shows the page for dog with ID 45.

- **Query String:**

`?energy=high` → Filters/extra data sent to the server (e.g., only show dogs with high energy)

- **Fragment:**

`#photos` → Jumps to the "photos" section on the profile page (client-side only).

Examples

- `https://pupperazzi.com/dogs/45/profile` Profile page for dog with ID 45.
- `https://pupperazzi.com/match?dogId=45&zip=14221` Finds nearby matches for dog 45 in ZIP 14221.
- `https://pupperazzi.com/match/swipe` user swipes left/right on potential match.
- `https://pupperazzi.com/messages?userId=88` Shows inbox for user 88, listing all active conversations.
- `https://pupperazzi.com/messages/send` route for sending a message (sender ID, receiver ID, message body).

Useful PHP/JS function

PHP Functions

- **parse_url(\$url)**

```
$url = "https://pupperazzi.com/match?dogId=45&zip=14221";
```

```
$parts = parse_url($url);
```

```
// $parts["path"] = "/match"
```

```
// $parts["query"] = "dogId=45&zip=14221";
```

Takes a full URL string and breaks it into components (scheme, host, path, query, fragment). Useful when you need to separate the query string from the path or check which page a user is requesting.

- **http_build_query(\$array)**

```
$params = ["dogId" => 45, "zip" => 14221];
```

```
echo http_build_query($params);
```

```
// "dogId=45&zip=14221";
```

Converts an associative array into a properly formatted query string. Saves time instead of manually writing ?key=value&key2=value2.

- **\$_GET["param"]**

```
echo $_GET["zip"]; // prints 14221
```

Accesses query string values from a URL. If the user visits /match?zip=14221, then \$_GET["zip"] returns 14221.

- **\$_POST["field"]**

```
echo $_POST["message"]; // prints the message text
```

Accesses data sent in a POST request (e.g., form submissions, API calls). Used for secure actions like sending a message or creating a new dog profile.

JavaScript Functions

- **new URL(urlString)**

```
let url = new URL("https://pupperazzi.com/messages?userId=88");  
console.log(url.pathname); // "/messages" console.log(url.searchParams.get("userId")); //  
"88";
```

Creates a URL object from a string so you can easily grab parts of it (like pathname or searchParams).

- **URLSearchParams**

```
let params = new URLSearchParams({ dogId: 45, zip: 14221 });  
console.log(params.toString()); // "dogId=45&zip=14221";
```

Helps build, read, and update query strings without string concatenation.

- **fetch(url, options)**

```
fetch("https://api.pupperazzi.com/messages/send", {  
  
  method: "POST",  
  
  body: JSON.stringify({ sender: 1, receiver: 2, text: "Woof!" }),  
  
  headers: { "Content-Type": "application/json" }  
  
})  
  
.then(res => res.json())  
  
.then(data => console.log(data));
```

Sends HTTP requests from the browser or Node.js. Can make GET (fetch data) or POST (send data) requests.

- **express.Router() (Node/Express)**

```
app.post("/messages/send", (req, res) => {
```

```
res.json({ status: "Message sent!" });
```

```
});
```

Organizes backend routes in a Node.js/Express server. Each route listens for requests and responds with data or HTML.